

The operating model

Durable local memory with explicit authority boundaries

OPERATING PRINCIPLE

SYNAPSE-S2 stores what was deliberately captured. It never claims unseen thoughts, files, or app state.

MEMORY ANATOMY

- Galaxy: every observed namespace and reviewed connection.
- Cortex: one namespace and its governed work state.
- Ganglion: a derived semantic or type cluster, not a stored object.
- Neuron: one durable memory entry with provenance and relationships.

AUTHORITATIVE CORE

- One authenticated local core owns SQLite writes, neural state, capture, recovery, and replication.
- CLI, MCP, and dashboard clients use that authority instead of fallback databases.
- The loopback dashboard is an operator surface, not a second source of truth.

WHAT PERSISTS

- Text, events, typed relationships, namespace catalog state, goals, and governed traces.
- Private image derivatives and their exact database references when image memory is used.
- Delivery, capture, recovery, replication, and governance receipts needed for audit.

WHAT STAYS TRANSIENT

- Graph layouts, open drawers, browser navigation, and other presentation state.
- Quick Prune and Deep Sleep maintain bounded runtime state; neither silently deletes durable memory.
- Unreviewed suggestions never become execution authority.

The Daily Operator Trust Loop

Establish current truth before relying on memory or starting risky work

OPERATING PRINCIPLE

Choose context -> Start Work -> Health and Doctor -> Enter Cortex -> Capture or Recall -> Wrap and protect.

START WORK

- Loads a bounded morning brief for the selected namespace.
- Shows recent evidence, active goals, risks, next actions, and leased delivery receipts.
- Acknowledgement happens only after content is successfully rendered or consumed.

GOVERNED WORK

- Enter Cortex with agent, mode, and task before substantial mutation.
- Tick before risky actions and declare intended files, tools, mutation intent, and confidence.
- Commit verified traces, then close instead of leaving stale active work.

TRUST CHECKS

- Context Health explains memory quality and recommendations.
- Doctor checks runtime, SQLite, embeddings, capture, delivery, and App Connect.
- Readiness evidence is stronger than a green dashboard badge alone.

CLEAN HANDOFF

- Wrap Session records decisions, validation, blockers, and next action.
- Create a recovery point before broad repair, pruning, or transfer.
- Keep commit, publication, installation, and live readiness as separate claims.

Durable memory and capture

Explicit creation paths with exactly-once processing and provenance

OPERATING PRINCIPLE

Capture is deliberate: bounded source text receives identity, is committed once, and remains attributable.

CREATION PATHS

- Remember stores one trace; ingest segments structured events and relationships.
- Conversation capture enters through the exactly-once capture ledger.
- App Connect previews locally exposed text before confirmed snapshot capture.

TYPED CONTEXT

- Goal, Decision, Risk, Validation, and Follow-up prefixes create typed context where appropriate.
- The active dashboard or CLI context determines the durable namespace target.
- Provenance remains available for recall, correction, and recovery checks.

EXACTLY-ONCE POSTURE

- Stable request identity and journals fail closed around uncertain outcomes.
- The inbox defers work while the core is unavailable, then resumes bounded batches.
- Credential-shaped material is rejected or redacted before durable ingestion.

CORRECTION

- Capture a replacement, then prune only the bad node or relationship.
- Arbitrary stored text is not edited in place.
- Broad relationship-class deletion is reserved for an entirely bad class.

Recall and Retrieval associations

Deterministic retrieval with visible evidence and bounded graph assistance

OPERATING PRINCIPLE

Retrieval associations are relevance cues, not user actions, approvals, tasks, or execution instructions.

RECALL SCOPES

- Local searches the active namespace plus explicit global memory.
- Connected adds approved, enabled, one-hop namespace bridges.
- All explicitly searches every saved namespace for that read-only request.

ASSOCIATIONS

- Derived terms can route a query toward related durable evidence.
- They stay subordinate to source memories and can be audited or revoked through Memora.
- Association output must never be described as operator intent.

RETRIEVAL V2

- Local neural embeddings and deterministic ranking return bounded evidence.
- Typed and graph-neighbor evidence can improve relevance without mutating memory.
- Recall cards expose provenance and why-matched detail for review.

BOUNDED NONCLAIMS

- A non-empty result does not prove relevance, truth, completeness, or success.
- Visible degree describes only the returned relationship sample.
- Benchmarks do not prove every real workload or every target Mac.

Namespace Galaxy and bridges

Read-only navigation plus reviewed cross-namespace recall

OPERATING PRINCIPLE

Galaxy -> Cortex -> Ganglion -> Neuron is semantic drill-down. Focus never edits, copies, or reconnects memory.

NAVIGATE

- Enter namespace changes the active operating scope.
- Scroll changes semantic depth; focus loads bounded neuron detail.
- The accessible list mirrors canvas navigation.

BRIDGE LIFECYCLE

- Suggestion -> pending proposal -> exact review -> approved enabled bridge.
- Disable, revoke, and expiry close recall authority without copying memory.
- Changed evidence invalidates stale review material.

VISUAL MEANING

- Galaxy area combines memory volume, indexed density, and enabled approved bridge weight.
- Ganglion and neuron size summarize bounded structure, not truth or importance.
- Suggestions and phase-delay values are presentation evidence only.

ISOLATION

- Connected recall follows approved one-hop paths and remains read-only.
- Suggestions never expand recall scope or affect durable state.
- The namespace catalog is not an ACL or ownership registry.

Private image and media memory

Locally derived visual evidence with explicit capture and recovery boundaries

OPERATING PRINCIPLE

Image memory is opt-in and local: thumbnail, description, and bounded descriptors support inspection and similarity.

CAPTURE

- The operator chooses an image and target namespace before creation.
- Transient source handling produces bounded private derivatives, not a hidden cloud upload.
- Explicit descriptions remain source-backed and inspectable.

SIMILARITY

- Find Similar resolves candidates from authoritative references.
- Distance is uncalibrated and carries truncation or exclusion notices.
- Feature bytes, raw OCR, and witness internals stay out of responses.

APPLE VISION LANE

- Local feature prints support image-to-image similarity.
- Consent-gated OCR contributes imperfect searchable cues.
- Neither lane is a VLM or calibrated text-to-image semantic search.

PRESERVATION

- Recovery and replication seal derivatives referenced by the immutable snapshot.
- Missing or corrupt referenced derivatives block publication.
- Valid orphan files are excluded and reported.

Memora governance

Compact abstractions and cues remain reviewable, attributable, and reversible

OPERATING PRINCIPLE

Memora improves relevance routing while source-backed durable memory remains authoritative.

SHADOW PLANNING

- A read-only plan proposes abstractions and cue anchors for one namespace.
- Planning never changes retrieval or writes an effective binding.
- Raw source text, vectors, supporting IDs, and witness internals stay hidden.

INTEGRITY

- Bindings, sources, provider identity, and receipts are cross-checked.
- Recovery and replication carry a content-free aggregate that recomputes exactly.
- Invalid source memories cannot stay effective through a stale cue.

GOVERNED LIFECYCLE

- Proposal records one isolated candidate for exact review.
- Promotion requires separated proposer and reviewer roles.
- Reject and revoke preserve append-only receipts and remove authority.

AUTHORITY BOUNDARY

- Cues influence retrieval relevance only.
- They do not create goals, tasks, approvals, follow-ups, bridges, or writes.
- Source memories remain independently inspectable and deletable.

Impact and evaluation

Observed behavior stays separate from illustrative estimates

OPERATING PRINCIPLE

Impact never claims provider billing, savings, relevance, correctness, or avoided work.

OBSERVED SCORECARD

- Counts non-empty dashboard recalls and bridge or graph-neighbor routing.
- Reports backend p50/p95 latency, response-byte token estimates, and warm coverage.
- Delivery ACK ratio is reliability evidence, not a quality score.

LONGMEM EVALUATION

- Bounded artifacts cover factual, temporal, update, abstention, and visual dimensions.
- The official adapter lane stays isolated from live operator memory.
- A benchmark gate supports comparison, not universal certification.

COST WHAT-IF

- The operator supplies a token rate and tokens-per-assist assumption.
- The dollar result is illustrative equivalence only.
- Coverage excludes MCP, CLI, and agent hydration traffic.

INTERPRETATION

- Non-empty recall is yield, not proven relevance.
- Latency covers backend retrieval, not complete client experience.
- Resource evidence is not external Instruments or Metal certification.

Cortex Governor and Goal Ledger

Typed work state preserves intent, evidence, risk, and next action

OPERATING PRINCIPLE

The governor disciplines work; it never replaces operator approval or turns relevance into authority.

SESSION LIFECYCLE

- Enter with agent, mode, and task.
- Tick with observation, proposed action, files, tools, mutation intent, and confidence.
- Commit typed evidence and close or wrap when the task ends.

GOAL LEDGER

- Goals carry owner, state, evidence, and next action.
- Active goals appear in Start Work and Cortex across CLI, MCP, and dashboard.
- A goal records coordination state; it does not authorize unrelated mutation.

TYPED TRACES

- Goal, decision, constraint, implementation, validation, risk, correction, and follow-up stay distinct.
- Truth posture, confidence, evidence, agent, and session identity remain attached.
- Promote and Demote apply to governed traces, not arbitrary memories.

GUARDRAILS

- Undeclared mutations, sensitive paths, and high-impact tools produce warnings.
- Cross-process closure prevents resurrection of ended sessions.
- An idle Cortex is normal before work begins.

Delivery and operational integrity

Bounded leases and exact receipts protect agent context

OPERATING PRINCIPLE

Delivery is at-least-once with stable identity; consumers deduplicate and acknowledge exact receipts.

HYDRATION

- Agent Brief combines leased events, recall, graph, Cortex state, and goals.
- Startup hydration does not acknowledge unseen events.
- Compact output retains visible receipts or releases the lease.

FAILURE SEMANTICS

- Deterministic invalid requests finish terminally and are not replayed.
- Genuinely uncertain commit state stays outcome-unknown and is never auto-replayed.
- Only pre-connect failure may retry the same signed request once.

RECEIPT LIFECYCLE

- A receipt is acknowledged only after successful delivery.
- Expired work gets a new fenced receipt with stable delivery identity.
- Release and dead-letter remain explicit operations.

CAPACITY

- Response channels and scans are bounded to protect core and client context.
- Terminal rows age out; sustained throughput remains finite until pruning.
- Rich dashboard views stay separate from compact MCP projections.

Evidence, recovery, and multi-Mac

Preserve database, capture, media, and authority as one verified story

OPERATING PRINCIPLE

Use paired signed recovery or target-bound replication, never an ad hoc live SQLite copy.

RECOVERY POINT

- A verified pair seals the immutable database snapshot and capture state.
- Referenced media derivatives are included by exact reference and digest.
- Isolated restore proof runs before the artifact is trusted.

OFFLINE REPLICATION

- Checkpoints are signed, paired, capability-negotiated, and target-bound.
- The receiver proves an isolated restore before signed acknowledgement.
- Any future live adoption requires a separately governed procedure; promotion is not supported here.

EVIDENCE PACK

- A readiness report and pinned recovery bundle preserve one run's proof.
- Certification checks core, embeddings, write, recall, App Preview, wrap, and dashboard smoke.
- Evidence is time-bound and must be refreshed for a new handoff.

HANDOFF RULE

- Quiesce the writer before final checkpoint creation.
- Current replication stops at isolated stage and signed ACK with `promotion_supported` false.
- Any later live use or return transfer needs separate governance; never improvise a two-way merge.

Safe release lane

Strong primitives exist; a conventional executable product does not yet

OPERATING PRINCIPLE

Production-ready local runtime and downloadable executable product are separate milestones.

WORKING TODAY

- A verified checkout runs the core, dashboard, CLI, MCP, capture, recovery, and offline replication.
- Source inspection, preservation, compatibility, and inactive staging exist.
- Readiness and replacement gates prove the installed checkout before launch.

NOT YET CLAIMED

- No signed and notarized macOS app or package is published.
- No offline installer, automatic migration, or executable rollback exists.
- No CI-backed multi-host canary proves unattended adoption.

DORMANT SUBSTRATE

- Provenance, layout, environment evidence, staging, and activation journals fail closed.
- They verify authority but do not compose one unattended cutover executor.
- Current evidence profiles do not form complete activation authority.

SAFE ORDER

- Verify provenance and compatibility; preserve state; stage immutably.
- Journal cutover, prove environment and equivalence, then commit the floor.
- Reconcile or roll back through evidence, never manual file replacement.

Capability boundaries and checklist

Treat unsupported capabilities as unavailable, never implied

OPERATING PRINCIPLE

SYNAPSE-S2 is production-usable local memory, not yet an unattended downloadable multi-host product.

MEMORY BOUNDARIES

- No in-place text or ganglion editor and no neuron reassignment.
- No namespace rename, archive, or delete lifecycle.
- No freeform relationship topology or weight editor.

BEFORE HOST HANDOFF

- Drain capture and establish one active writer.
- Create a signed target-bound checkpoint and use encryption.
- Prove isolated restore before ACK; treat live adoption as unsupported until separately governed.

INTELLIGENCE BOUNDARIES

- Associations and Memora cues route relevance only.
- Vision features are not a VLM or identity system.
- Impact and benchmarks do not prove truth or savings.

BEFORE RELEASE CLAIMS

- Verify the exact commit independently on every remote.
- Separate publication, installation, launch, and live readiness.
- Name missing installer, migration, rollback, and canary work plainly.